

Yuchen Wang

Agentic post-training | ML systems

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Education

University of Illinois Urbana-Champaign	2025 – 2027 (expected)
M.S. in Computer Science	Champaign, IL
Peking University	2021 – 2025
B.S. in Intelligent Science and Technology (AI); Zhi Class (Elite Program), Top 10%	Beijing, China

Research Experience

Alibaba U.S. – Accio Team	May 2026 – Dec 2026
Research Intern, Agentic LLM Post-Training	Sunnyvale, CA
- Core contributor to Occamy-1.0 [arXiv · Website], a 35B-A3B agent model. Built execution-grounded data and post-training infrastructure; the team's SFT corpus contains 15K trajectories and 403M tokens. - Developed verifier-gated pipelines for OpenClaw, Hermes, and Accio Work: token-exact replay, state recovery, and compaction-aware segments preserving tokens, log-probs, and loss masks for SFT/RL. - The released model improves Claw-Eval average from 69.5 to 82.2 and AutomationBench strict pass from 7.5% to 27.6%, reaching the Pareto frontier in the report's four-benchmark cost comparison.	
Freedo Technology	May – Aug 2025
Research Intern, 3D Vision and Generative AI	Beijing, China
Built depth/normal-conditioned ControlNet for 3D building reconstruction from noisy point clouds, preserving structural edges and planar surfaces; improved geometric fidelity from 32.7% to 85.3%.	

Open-Source Systems Engineering

- Inference performance.** [vLLM](#): halved BF16 MoE conversion peak allocation (TP2, **7.88 to 3.94 GiB**). [NeMo RL](#): deferred teacher top-k payloads for **4.44–5.31× speedup** in controlled Ray transfer measurements.
- Numerics and long-context training.** Fixed [FlashInfer FP8 KV calibration](#) and [small-norm Muon normalization](#); fused [GatedDeltaNet Q/K normalization](#) and added [selective Mamba recompute](#) in Megatron-LM. Proposed the adopted [FlashAttention backward JIT-key fix](#) without CPU–GPU sync.
- Serving and agent reliability.** Fixed [Mamba cache-publication ordering](#) under SGLang overlap scheduling. Repaired Hermes [schema coercion](#) and [SSH reconnect](#); added [partial-stream continuation controls](#). Co-authored [pickle-safe HTTP error propagation](#) in NeMo Gym.

Selected Research Projects

CineFlow: Dependency-Driven Parallel Video Generation [Paper]	Oct 2025 – May 2026
Advisor: Prof. Fan Lai, UIUC	
- Built semantic DAGs, runtime trajectory fusion, online edge pruning, and critical-path scheduling for video diffusion without retraining; analyzed dependency structure in 1K+ MovieGenBench prompts. - Across Wan2.2-5B, CogVideoX-5B, and HunyuanVideo on 8×H100: 1.7–5.5× end-to-end speedup, 1.30–2.02× lower P90 latency, and 5.4–17.3% higher VBench overall.	
Dynamic Prefill Optimization via Adaptive Online Packing [Report]	Aug 2025 – Jan 2026
University of Illinois Urbana-Champaign	
Built AIMD batching control using live p95 TTFT feedback and greedy/DP length-aware packing; reduced TTFT by up to 20% on production-style DynamoLLM traces.	
FlashAttention-style CUDA Kernels for GPT-2 [Report]	Aug – Dec 2025
University of Illinois Urbana-Champaign	
Implemented tiled attention, online softmax, and kernel fusion: arithmetic intensity 15 to 157 FLOP/byte, roughly 10× lower HBM traffic, and up to 9% end-to-end speedup.	
Reinforcement Learning for Legal Reasoning [Thesis]	Jan – Mar 2025
Advisor: Prof. Yansong Feng, Peking University	
Built a Zero-RL → distilled-CoT SFT → GRPO pipeline; reached 57.6% accuracy, outperforming SFT-only and RL-only baselines.	

Technical Skills

Languages: Python, C++, CUDA. **Systems:** PyTorch, Ray, FSDP, Megatron-LM, NeMo RL, vLLM, SGLang, Nsight.
Methods: RL, distributed training, diffusion transformers, ControlNet, 3D vision and point clouds.